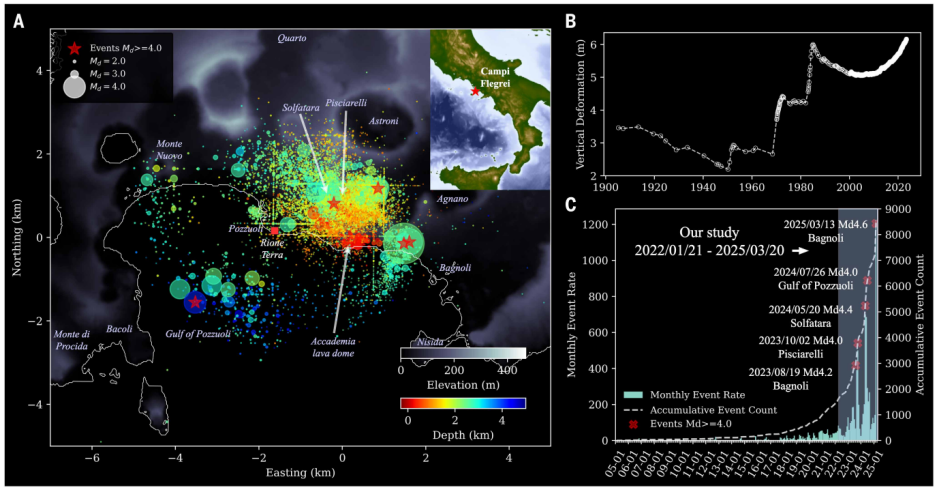
Figure 1: Seismicity and ground displacement

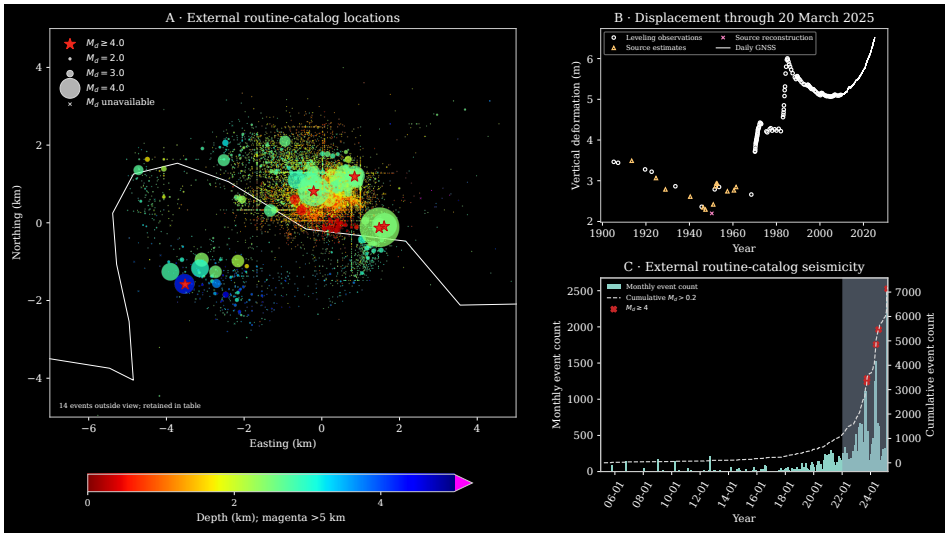
Published reference



Tan et al. (2025), tanmodel.pdf, p. 2.
DOI: 10.1126/science.adw9038.

Independent result

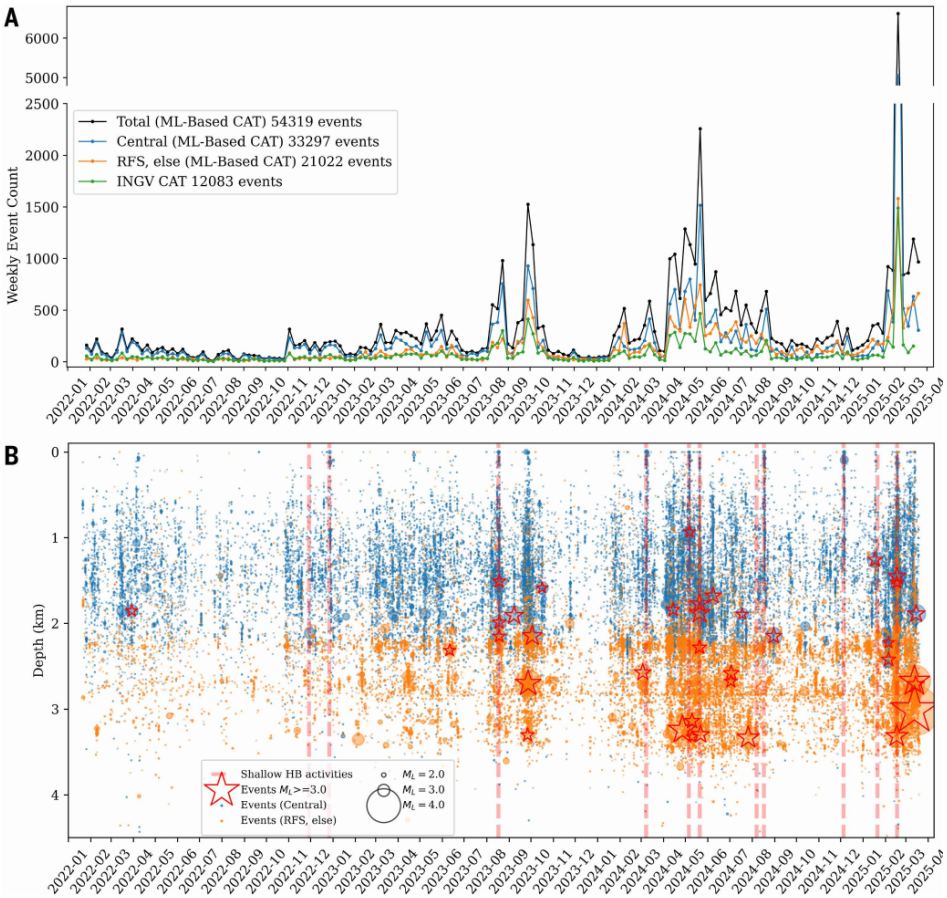
Panel 1A–C — partially reproduced



Partial contextual reconstruction. A: external routine catalog, 16,046 located events and five Md>=4 stars. B: 114 leveling surveys (circles), 13 source-provided estimates (triangles), one source-provided reconstruction (cross), and daily GNSS through 20 March 2025; Historical estimates are not raw observations or reconstructions made here. C: all routine monthly counts and cumulative Md>0.2; March includes days 1–20 only. Routine events are not independent detections. Datum joining is an assumption; basemap and live snapshot differ from the publication. Depth colors: fixed 0âĂŞ5 km, red to blue; magenta >5 km, no data clipping. S5 limits/ticks are printed; 1A endpoints are unlabeled and 0âĂŞ5 km is an explicit approximation. Circle area law is an explicit display assumption, not recovered exactly. Missing magnitudes are crosses. Routine events are external context.

Figure 2: Temporal evolution of seismicity

Published reference



Tan et al. (2025), tanmodel.pdf, p. 3.
DOI: 10.1126/science.adw9038.

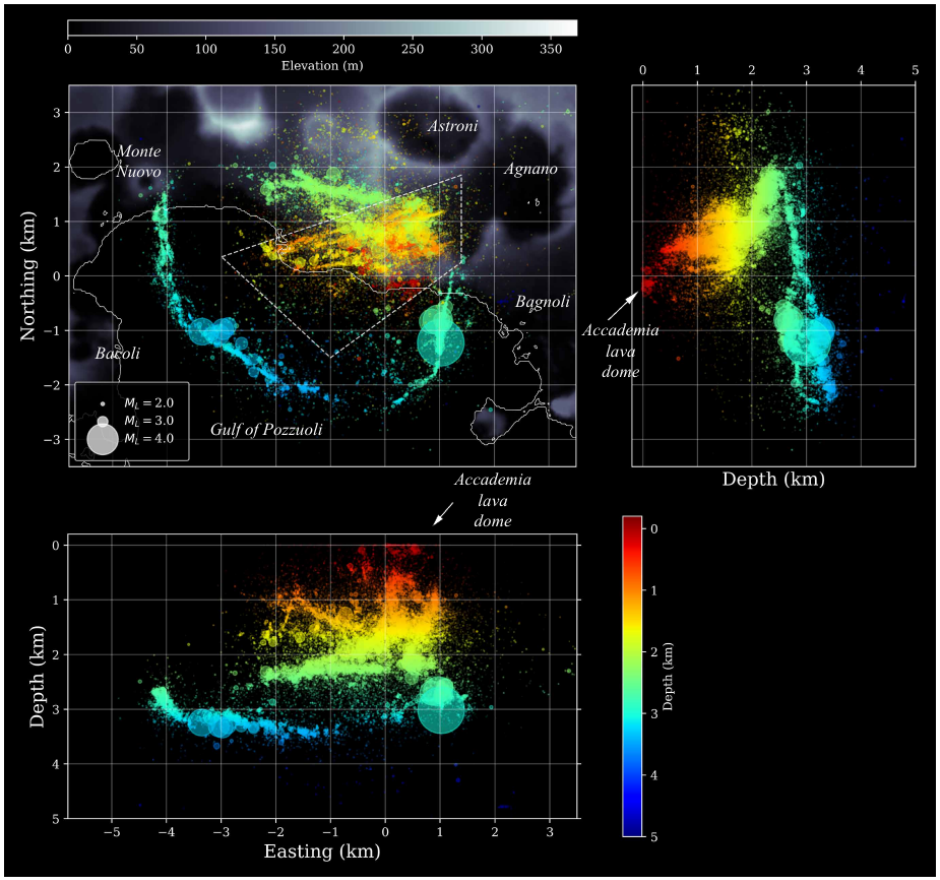
Independent result

Missing: 2A, 2B — not reproduced

Panels A and B need a validated multiyear relocated catalog and independent magnitudes. The expanded 1,015 associations now enter grid-derived calibration and two TomoDD trials, but path sensitivity and sparse coverage prevent acceptance.

Figure 3: Spatial distribution of seismicity

Published reference



Tan et al. (2025), tanmodel.pdf, p. 4.
DOI: 10.1126/science.adw9038.

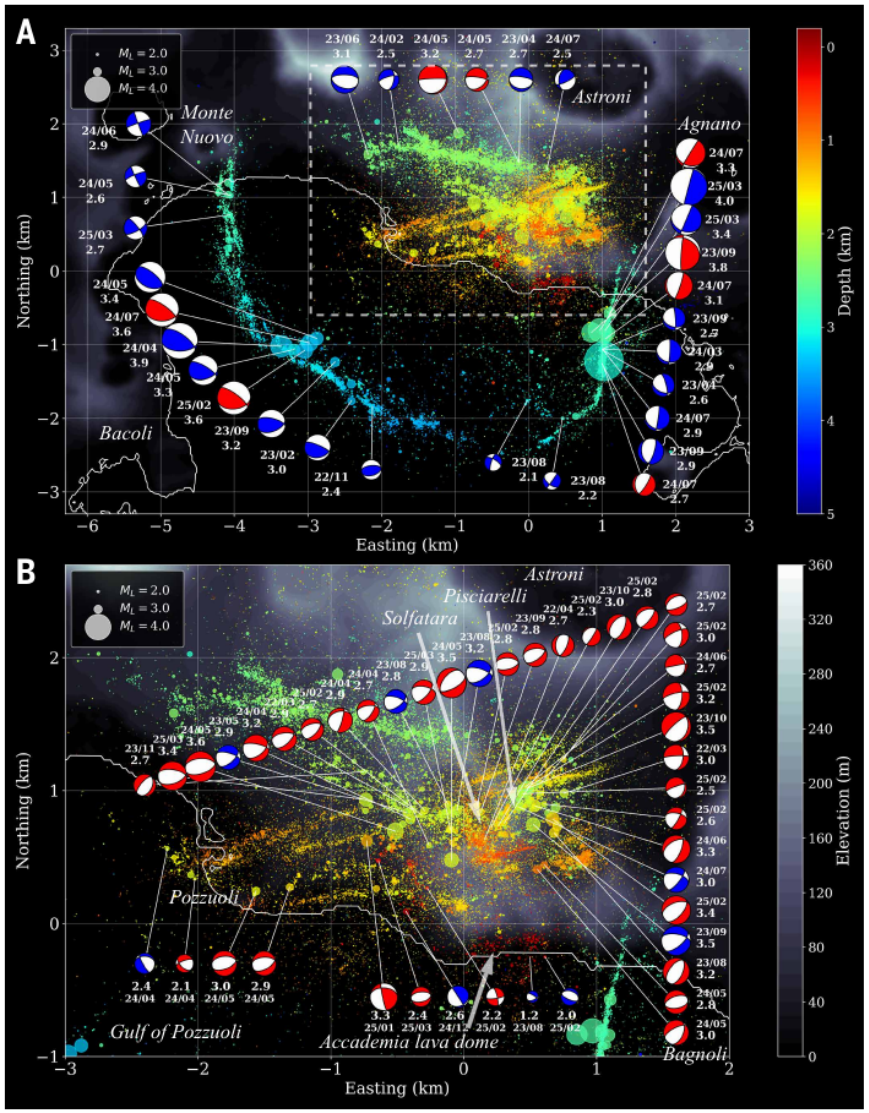
Independent result

Missing: 3/map, 3/northing-depth, 3/easting-depth — not reproduced

The map and both cross sections require independently calibrated and double-difference-relocated events. Provisional diagnostic locations are not substitutes.

Figure 4: Focal mechanisms of larger events

Published reference



Tan et al. (2025), tanmodel.pdf, p. 5.
DOI: 10.1126/science.adw9038.

Independent result

Missing: 4A, 4B — not reproduced

Panels A and B require independent SKHASH/gCAP solutions and reliable locations. No focal inversion was executed.

Figure S1: Recorded station availability

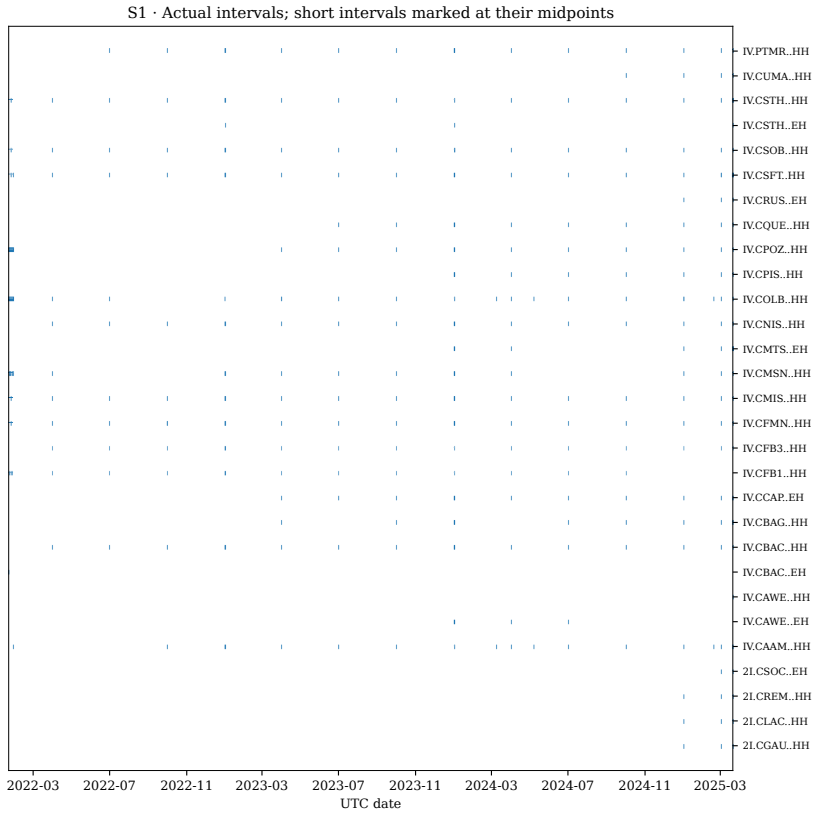
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 12.
DOI: 10.1126/science.adw9038.

Independent result

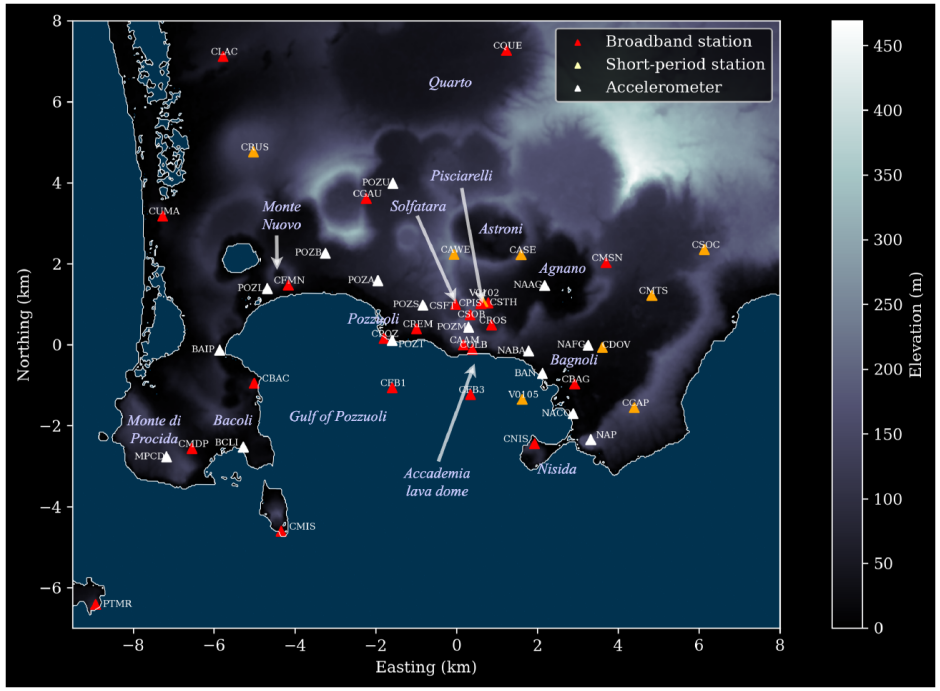
Panel S1 — partially reproduced



Partial: expanded checkpoint, 868 processed files, 26 station names, 8511.520 union channel-hours. Empty intervals are neither connected nor counted. Current grid-derived calibration uses this expanded checkpoint. Earlier diagnostics retain separate identifiers. Vertical ticks show observed-interval midpoints where durations are too short to resolve; ticks do not imply continuous coverage.

Figure S2: Station locations

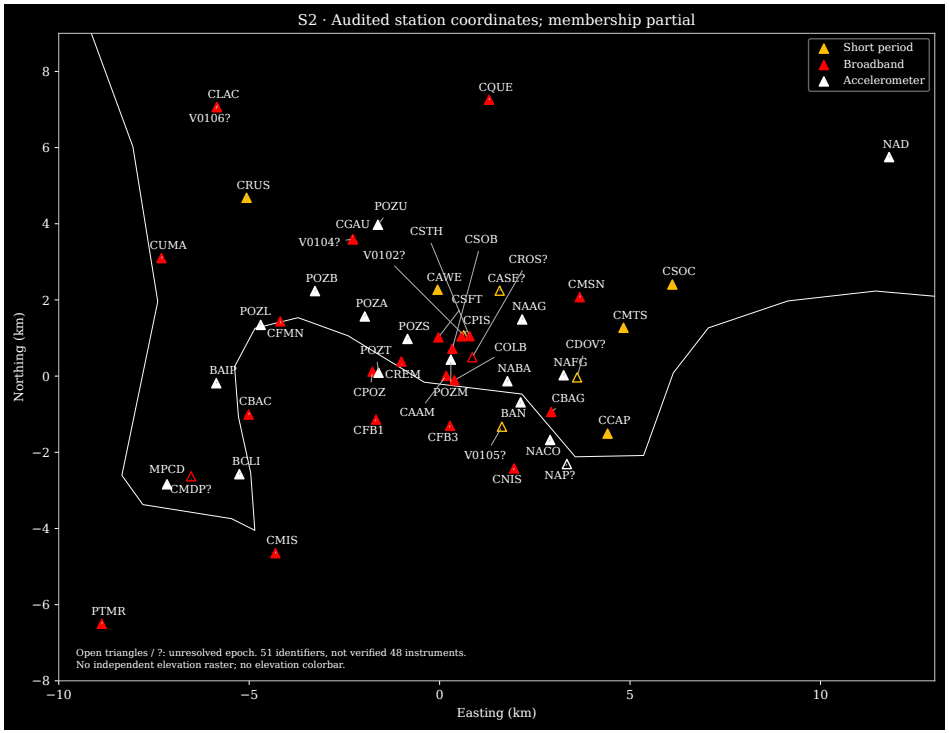
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 13.
DOI: 10.1126/science.adw9038.

Independent result

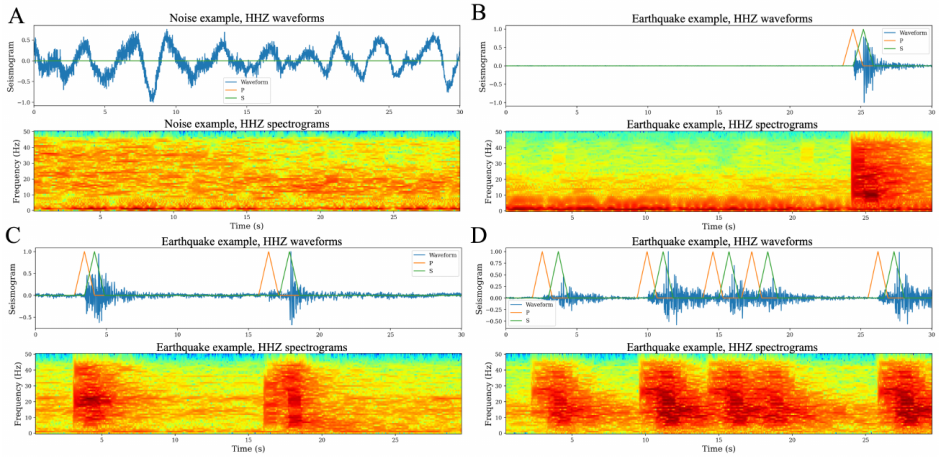
Panel S2 — partially reproduced



Partial station inventory: 51 released identifiers audited against official study-period response epochs. Filled triangles: 42 official coordinate matches; open triangles: 9 released coordinates with unresolved epochs or channel codes. Two coincident-coordinate pairs remain distinct identifiers. No verified geographic outlier. Local WGS84 projection and clipped coastline; the original 48-instrument membership is not established. Colors distinguish broadband (red), short period (yellow) and accelerometer (white) instruments; no independent relief raster is available.

Figure S3: Training examples

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 14.
DOI: 10.1126/science.adw9038.

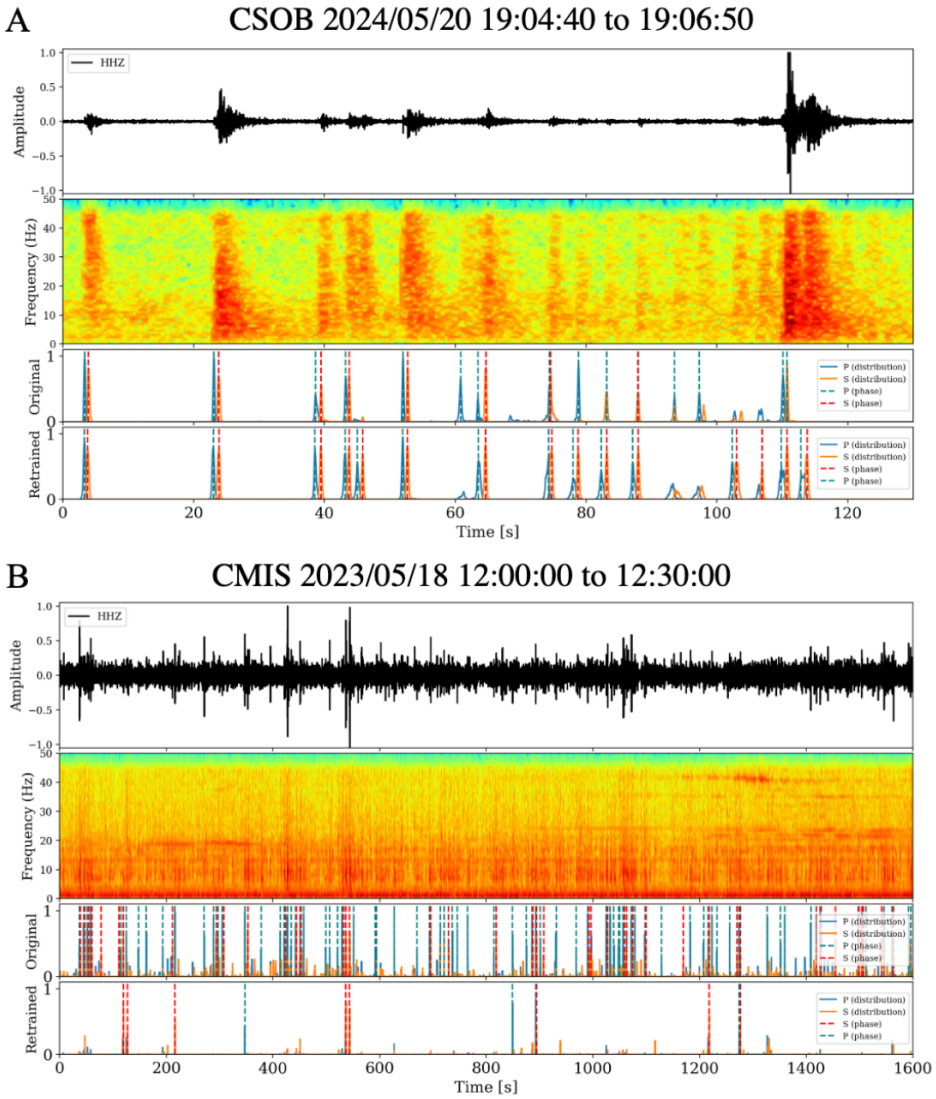
Independent result

Missing: S3A, S3B, S3C, S3D — not reproduced

Panels A–D require the specified training traces, annotations and augmentation examples. These were not reconstructed; downstream use of trained weights is not retraining.

Figure S4: PhaseNet benchmark examples

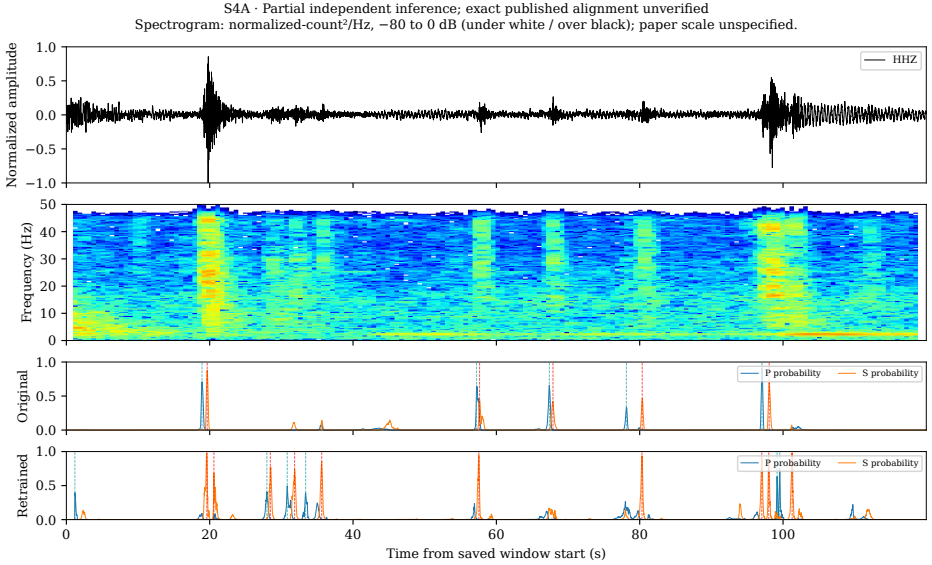
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 15.
DOI: 10.1126/science.adw9038.

Independent result

Panel S4A — partially reproduced



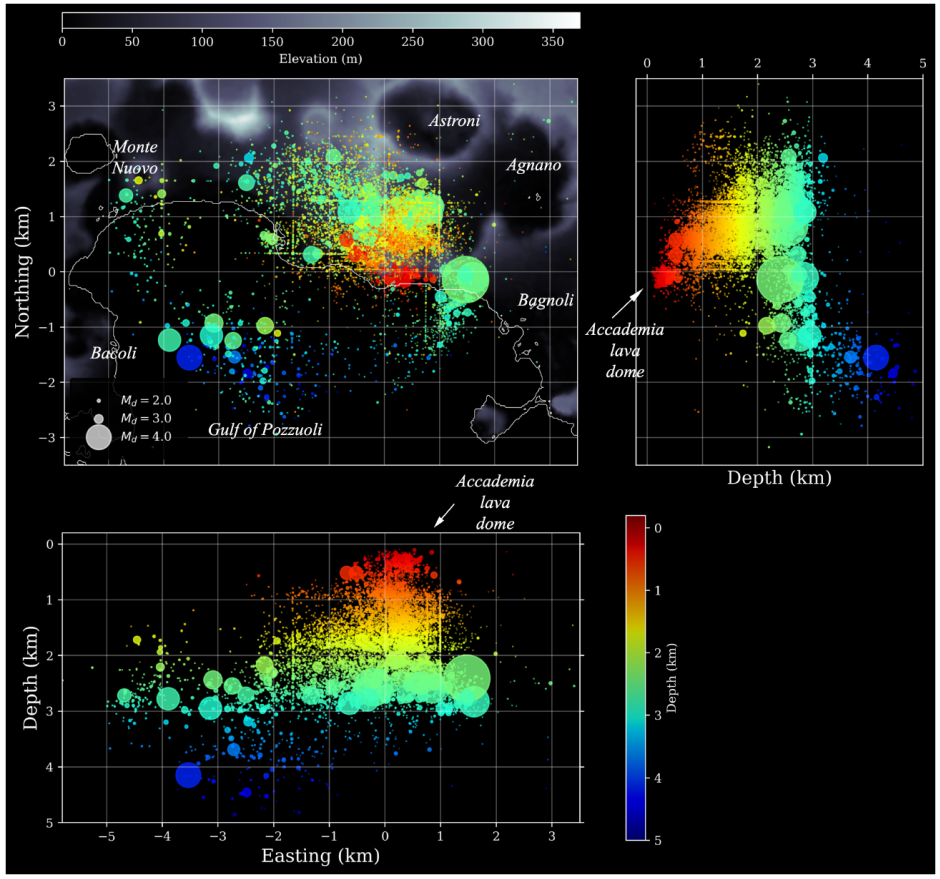
Partial S4A inference comparison on original CSOB observations, 20 May 2024 20:10:03–20:12:03 UTC. The caption's two-hour offset is interpreted literally; exact published alignment is unverified. Original weights: 5 P and 5 S; retrained weights: 6 P and 10 S. No labeled accuracy or retraining claim.

Missing: S4B — not reproduced

S4B: exact CMIS noise-window timing and reference labels are unavailable. Other calendar-window diagnostics are not inserted as that panel.

Figure S5: Routine-catalog locations

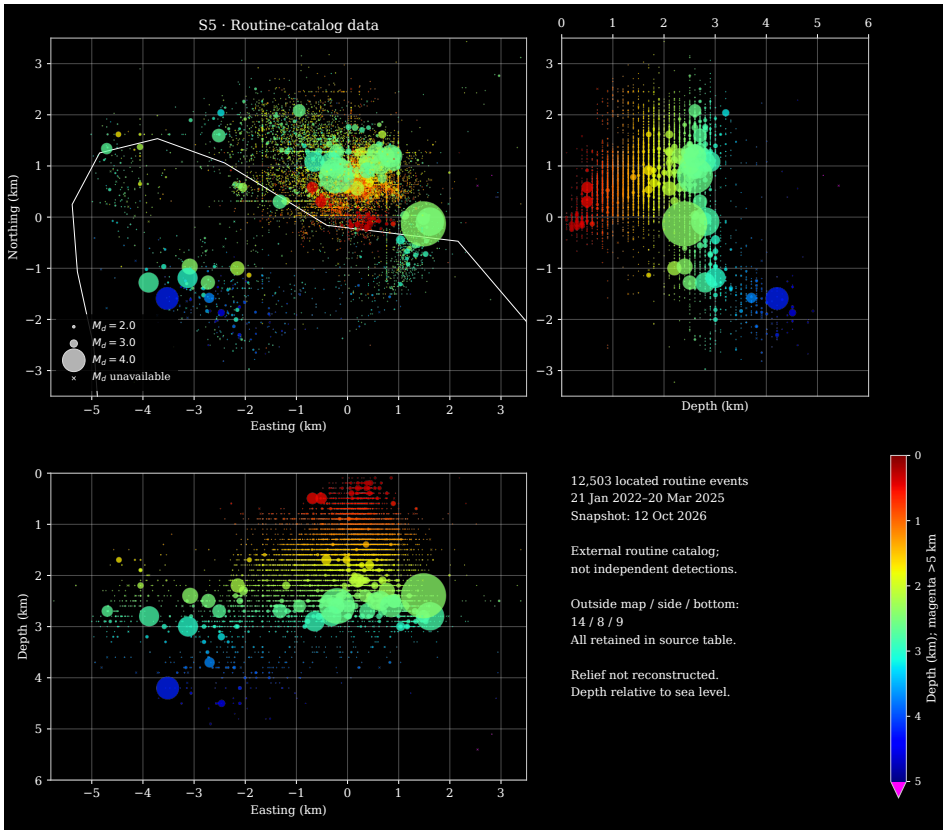
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 16.
DOI: 10.1126/science.adw9038.

Independent result

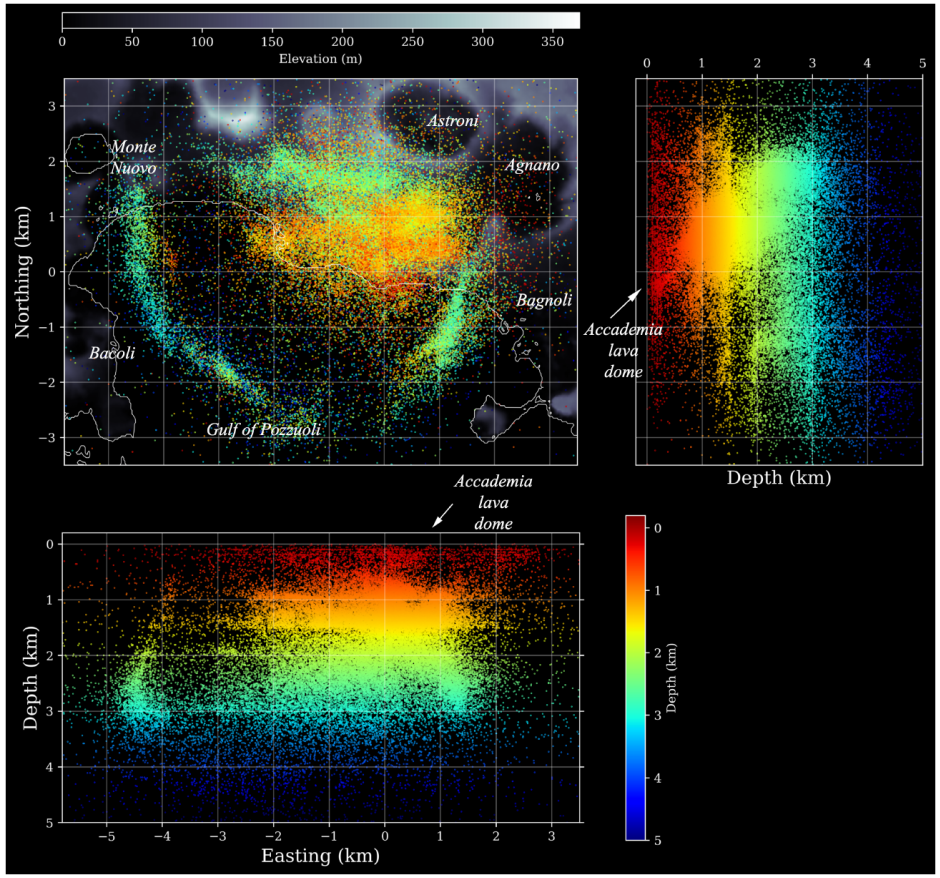
Panel S5, all views — partially reproduced



External routine-catalog data, independently selected and plotted: 12,503 located events, 21 January 2022–20 March 2025. This is 420 more than the published historical snapshot. No author-derived catalog is used to adjust that difference; these are not independent detections. Depth colors: fixed 0âĂŞ5 km, red to blue; magenta >5 km, no data clipping. S5 limits/ticks are printed; 1A endpoints are unlabeled and 0âĂŞ5 km is an explicit approximation. Circle area law is an explicit display assumption, not recovered exactly. Missing magnitudes are crosses. Routine events are external context.

Figure S6: Initial VELEST locations

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 17.
DOI: 10.1126/science.adw9038.

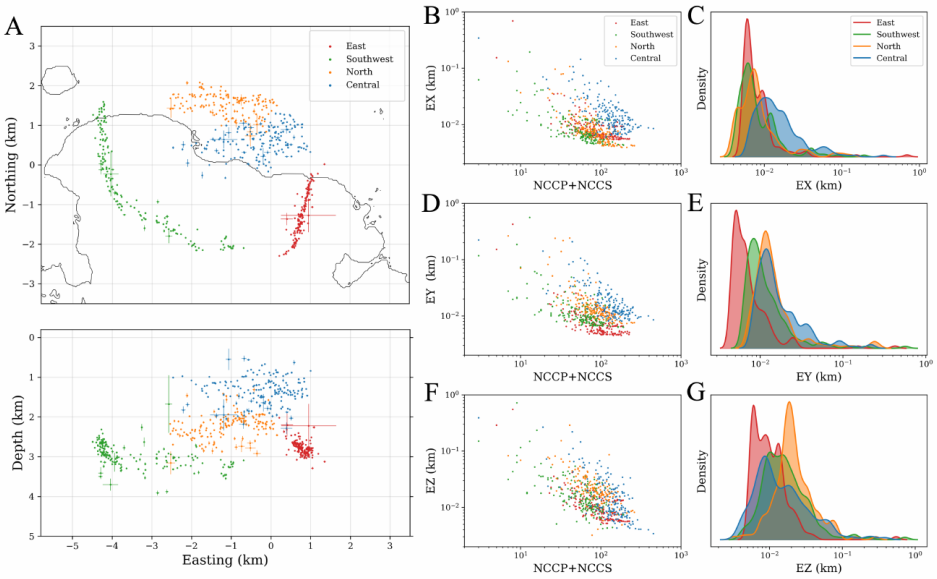
Independent result

Missing: S6/map, S6/northing-depth, S6/easting-depth — not reproduced

No validated independent counterpart. Authorized-grid VELEST now runs on the expanded cohort but remains path-sensitive; both fixed-grid rounds executed; validation pending. Diagnostics are retained after the supplement. No rejected locations are presented as reproduced structure.

Figure S7: Location uncertainties

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 18.
DOI: 10.1126/science.adw9038.

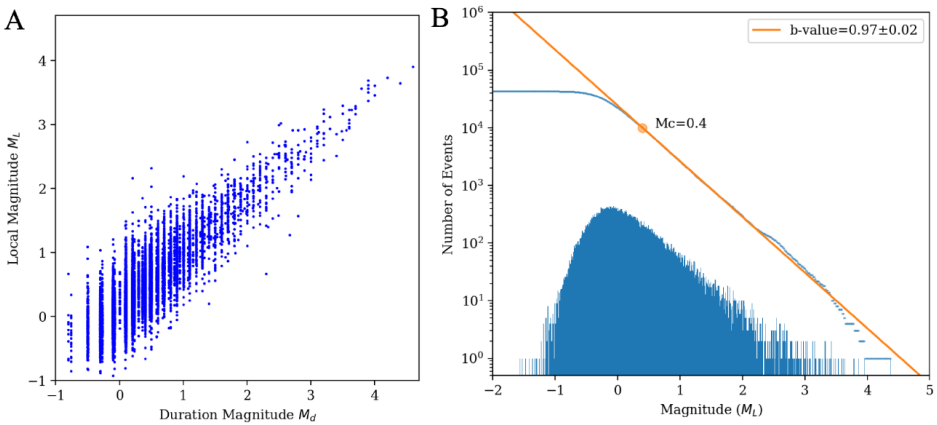
Independent result

Missing: S7A/map, S7A/easting-depth, S7B, S7D, S7F, S7C, S7E, S7G — not reproduced

No validated independent counterpart. Authorized-grid VELEST now runs on the expanded cohort but remains path-sensitive; both fixed-grid rounds executed; validation pending. Diagnostics are retained after the supplement. No rejected locations are presented as reproduced structure.

Figure S8: Magnitude calibration and frequency distribution

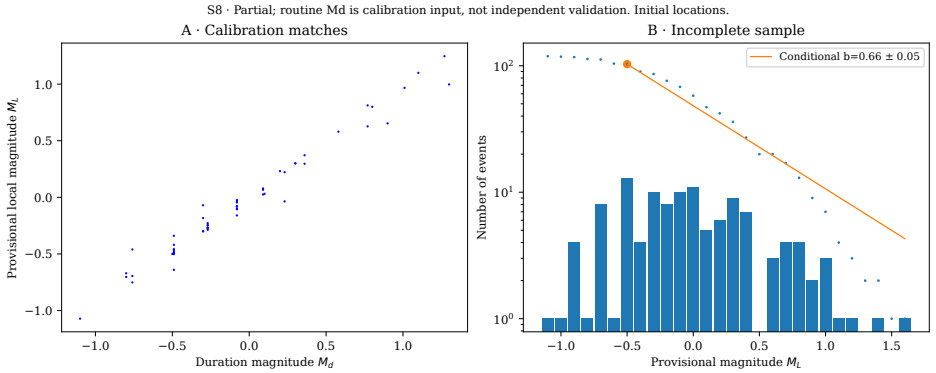
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 19.
DOI: 10.1126/science.adw9038.

Independent result

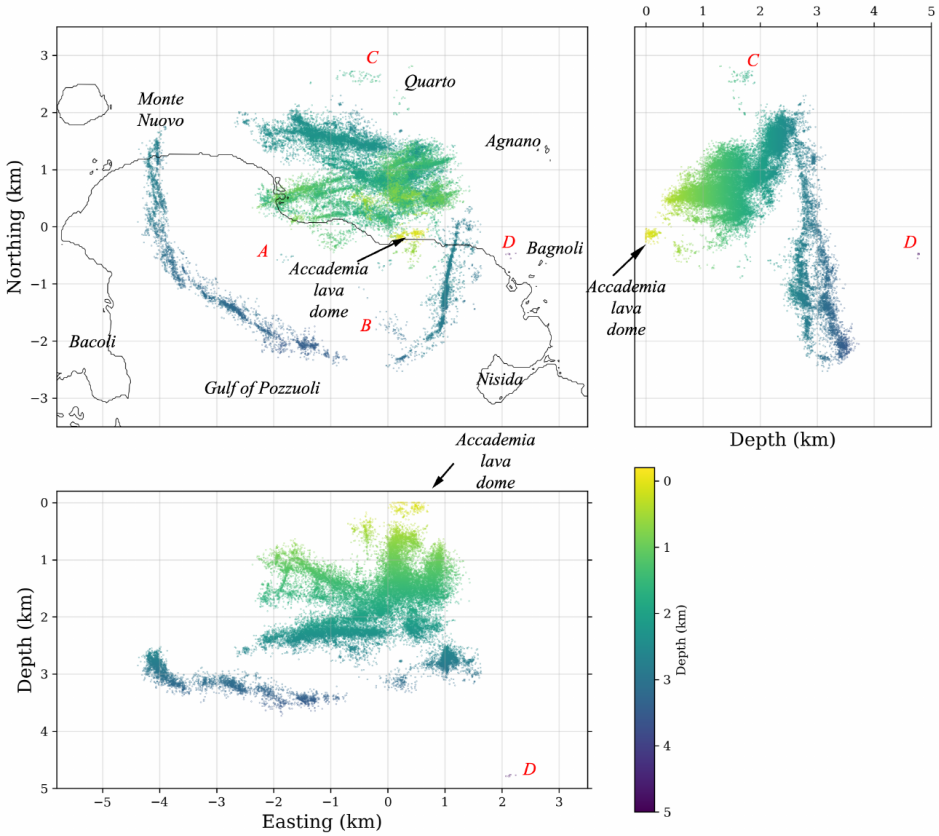
Panel S8A–B — partially reproduced



Partial amplitude-ratio pilot: 119 provisional magnitudes from independent observations, with routine Md as the documented calibration prior. A is not an independent validation of that prior. B is a limited-sample histogram with a conditional likelihood estimate, not the full-period completeness or b-value result. Locations are initial associations, before final relocation.

Figure S9: Best-located earthquake subset

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 20.
DOI: 10.1126/science.adw9038.

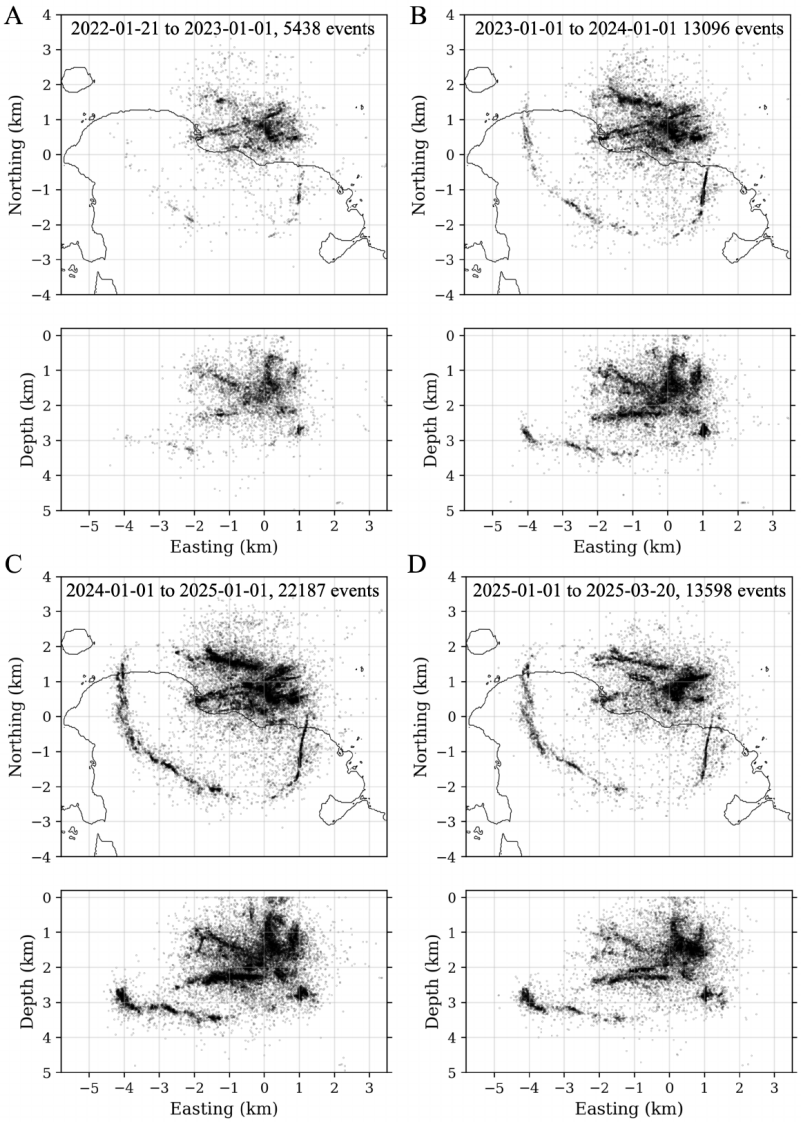
Independent result

Missing: S9/map, S9/northing-depth, S9/easting-depth — not reproduced

All three views require independently relocated events and cross-correlation observation counts for the published quality filter. These outputs are absent.

Figure S10: Seismicity by year

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 21.
DOI: 10.1126/science.adw9038.

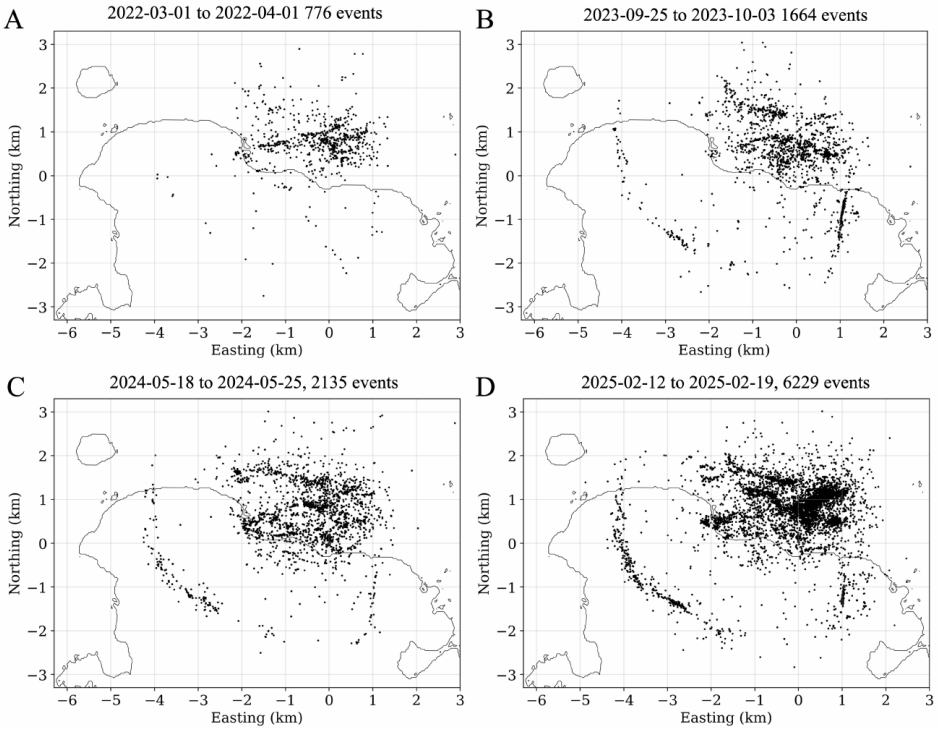
Independent result

Missing: S10/2022, S10/2023, S10/2024, S10/2025 — not reproduced

The four annual groups for 2022–2025 require the independent multi-year relocated catalog. The available sample does not support them.

Figure S11: Four major swarms

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 22.
DOI: 10.1126/science.adw9038.

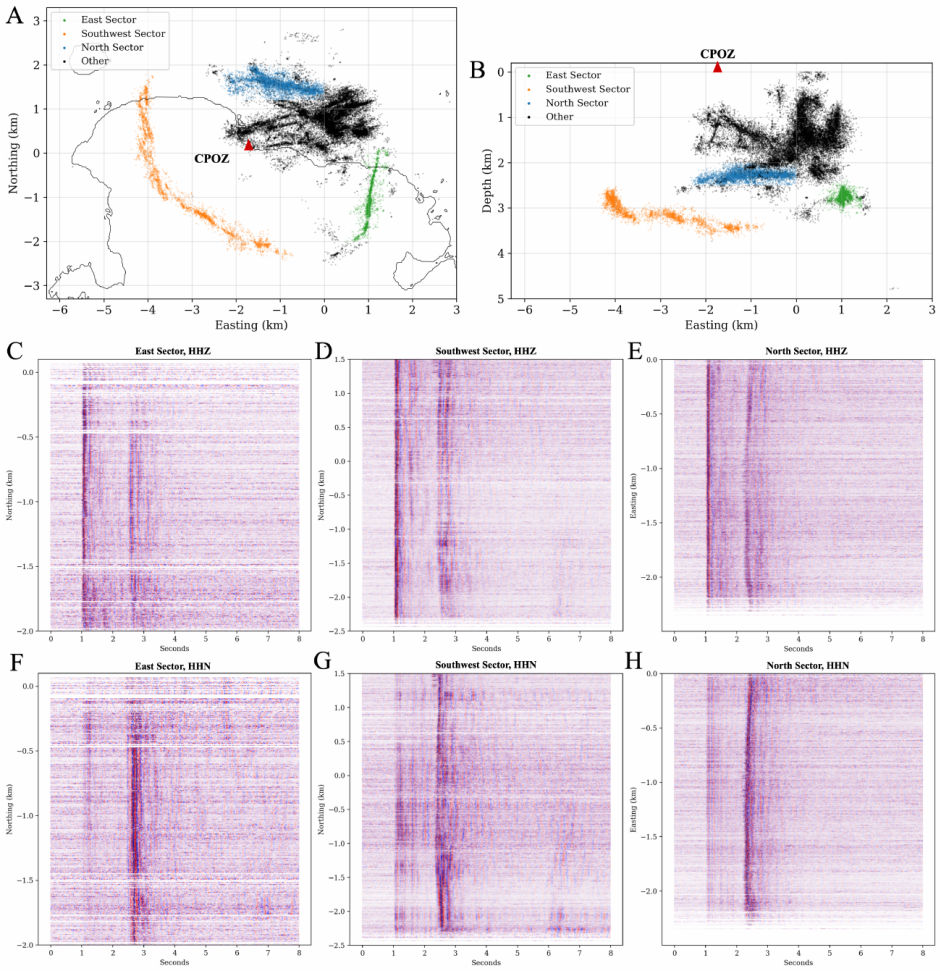
Independent result

Missing: S11/swarm-1, S11/swarm-2, S11/swarm-3, S11/swarm-4 — not reproduced

The four swarm maps require the independent relocated catalog and justified swarm windows. These selections and locations remain incomplete.

Figure S12: Ring sectors and CPOZ common-receiver gathers

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 23.
DOI: 10.1126/science.adw9038.

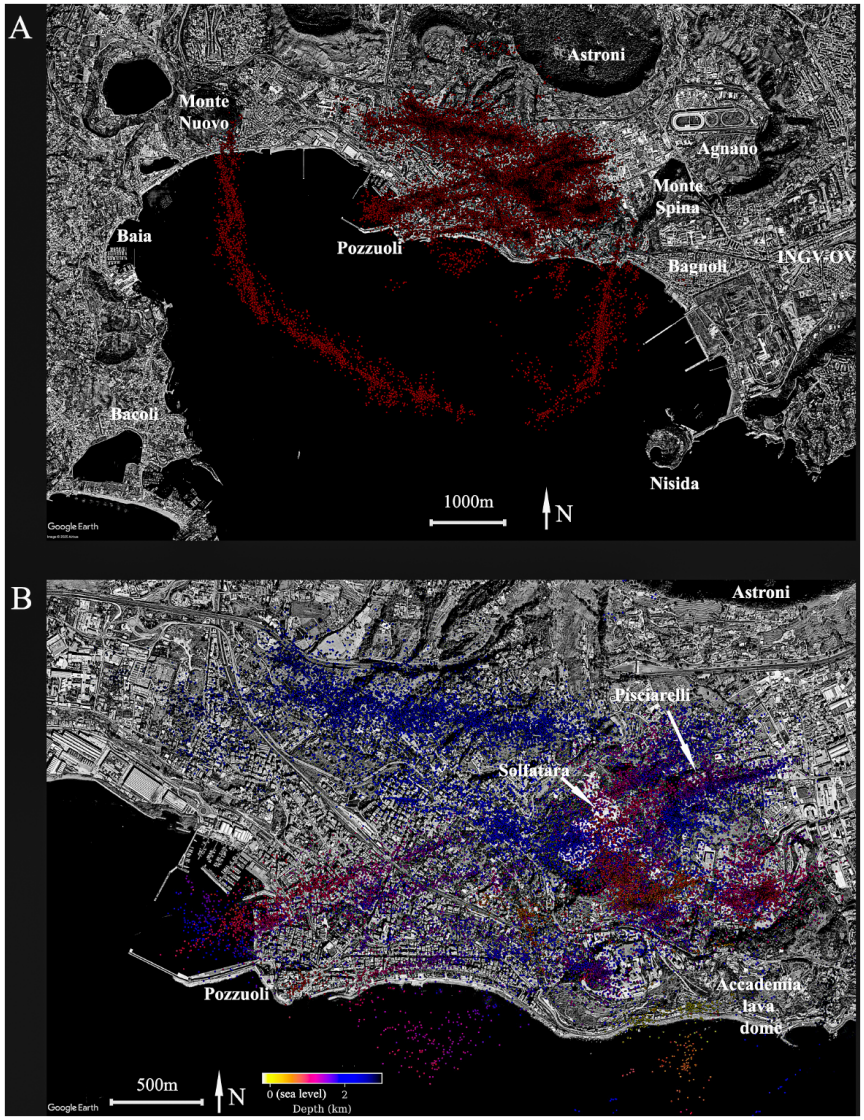
Independent result

Missing: S12A, S12B, S12C, S12D, S12E, S12F, S12G, S12H — not reproduced

Panels A–H need relocated sector selections and manual depth review. 137 independent CPOZ P arrivals now exist, but the required sector gathers remain unfinished.

Figure S13: Relocated seismicity on satellite imagery

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 24.
DOI: 10.1126/science.adw9038.

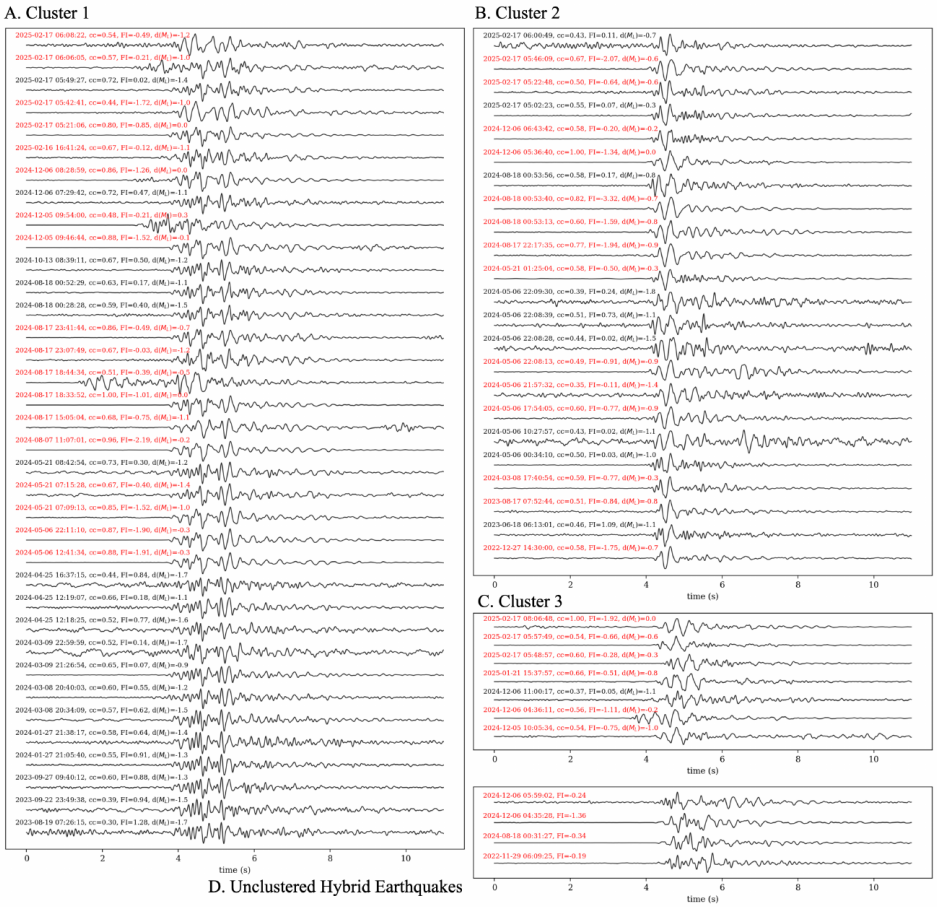
Independent result

Missing: S13A, S13B — not reproduced

Panels A and B require reliable independent relocated coordinates and an appropriately sourced basemap. The relocated catalog does not exist.

Figure S14: Hybrid-event clusters

Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 25.
DOI: 10.1126/science.adw9038.

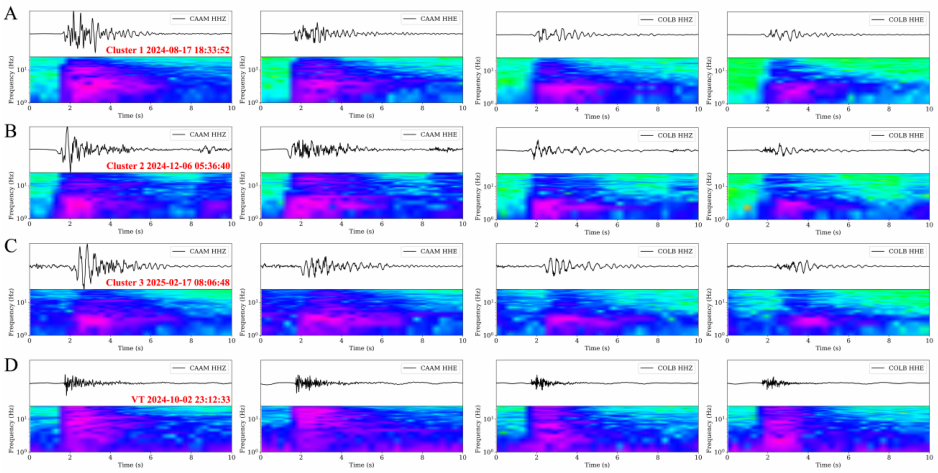
Independent result

Missing: S14/1, S14/2, S14/3, S14/unclustered — not reproduced

The cluster histories require three-year exposure, confirmed events and relative magnitudes. The present 75.797 hour scan produced 822 unconfirmed threshold crossings, not reviewed families.

Figure S15: Hybrid and shallow-event waveforms

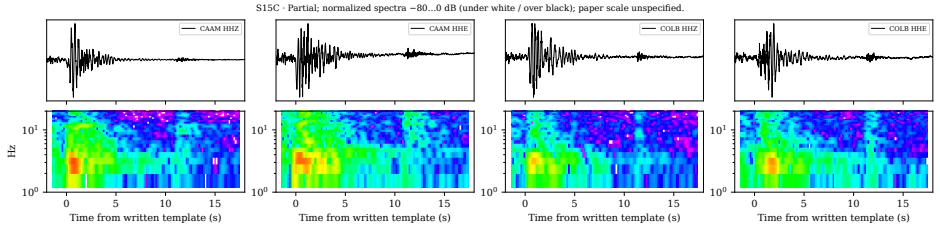
Published reference



Tan et al. (2025), tanmodelsupp.pdf, p. 26.
DOI: 10.1126/science.adw9038.

Independent result

Panel S15C — partially reproduced



Panel C is partial: independently processed E3 CAAM/COLB observations. Onset alignment, display filtering and spectral settings differ or are incompletely specified. A, B and D have placeholders rather than substitute waveforms.

Missing: S15A, S15B, S15D — not reproduced

Panels A and B use different published events from the saved E1/E2 templates; D has no independently selected matching example. Those panels remain not reproduced.